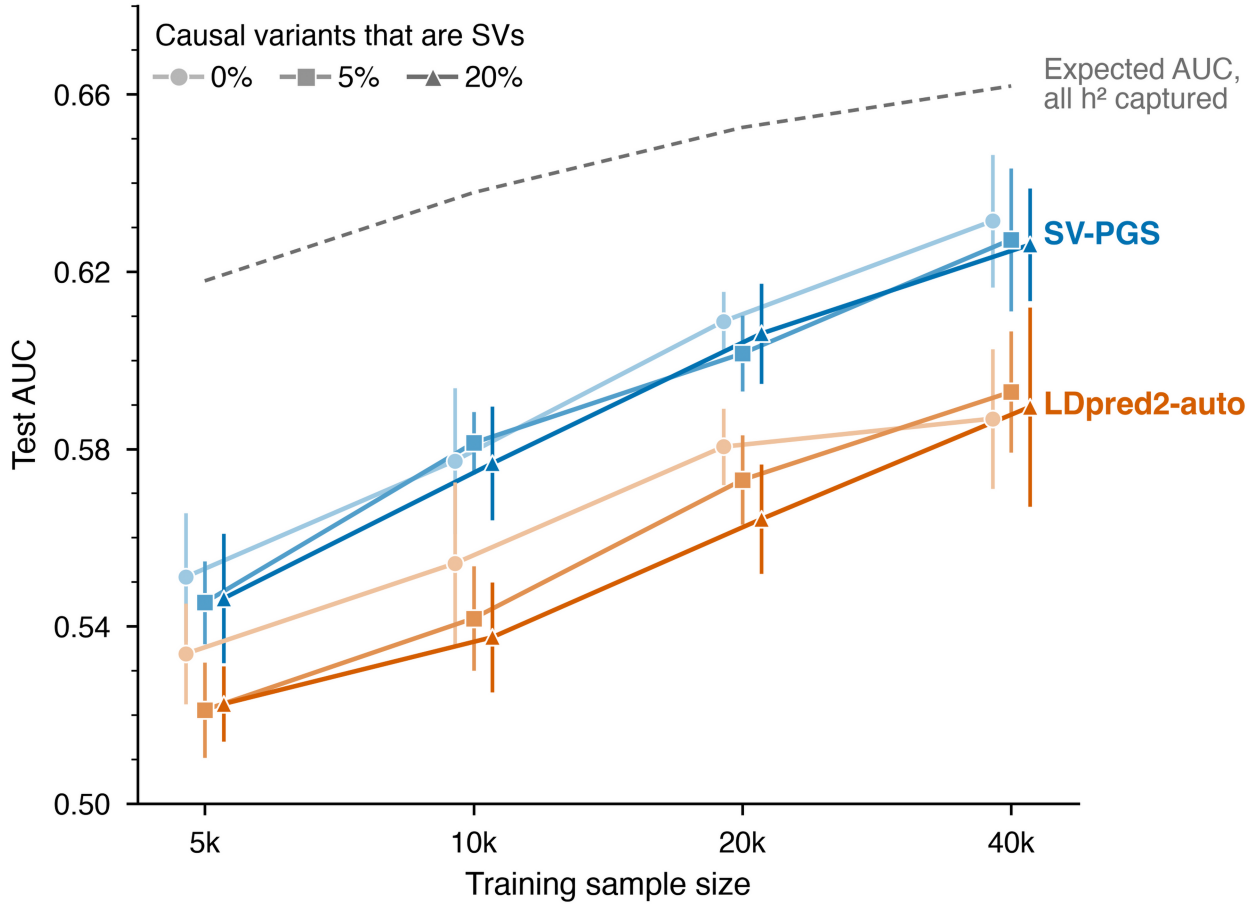


Figure 1 | SV-PGS gains accuracy faster than LDpred2-auto as the training sample grows, whatever the share of causal SVs. Test AUC in 10,000 held-out people for a simulated binary trait on chromosome 22 (liability $h^2 = 0.1$, prevalence 10%, 200 causal variants) in which 0%, 5% or 20% of the causal variants are SVs with the same per-variant effect distribution as small variants. SV-PGS fits every SNV, indel and imputed SV; LDpred2-auto fits HapMap3-density SNVs of the same training people. Points are means over simulation replicates (5-15 per point) with Student-t 95% intervals, offset horizontally for legibility; the dashed line is the expected AUC of a score capturing all of h^2 (Daetwyler et al. 2008; Wray et al. 2010). bench-sim v7 cohort from 1000 Genomes haplotypes, Beagle-imputed SVs; fitted on MSI.

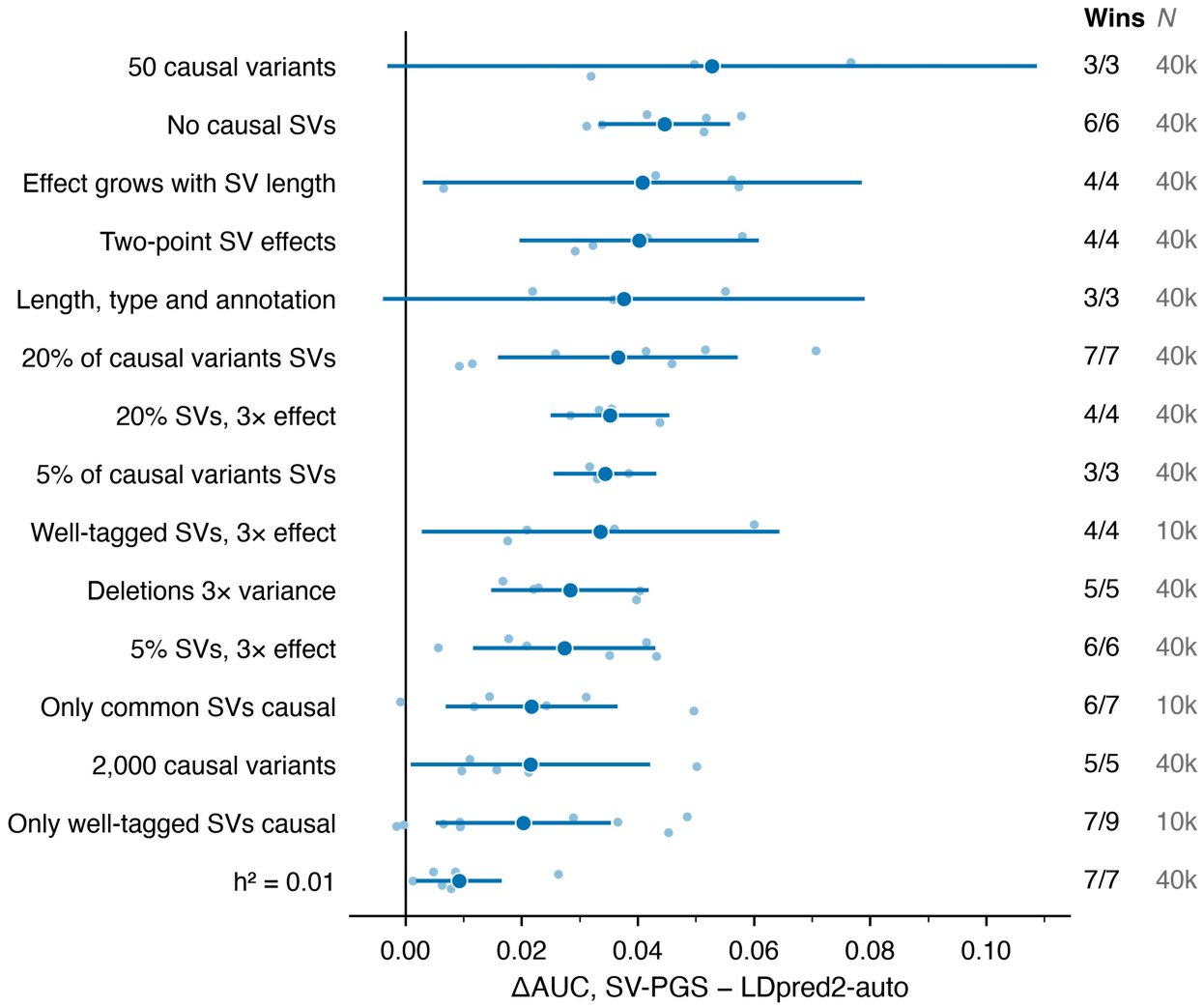


Figure 2 | SV-PGS is more accurate than LDpred2-auto across simulated genetic architectures. Paired difference in test AUC (same 10,000 held-out people) between SV-PGS with SVs and LDpred2-auto, per replicate (small dots), with the mean and its Student-t 95% interval, at the largest training size with at least three replicates (N, right). Wins: replicates in which SV-PGS had the higher AUC (74 of 77 overall). Unless named, each architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs; '3× effect' triples the SV effect variance, 'well-tagged' draws causal SVs only from those imputed with $r^2 \geq 0.8$, 'common' only from SVs of frequency ≥ 5%.

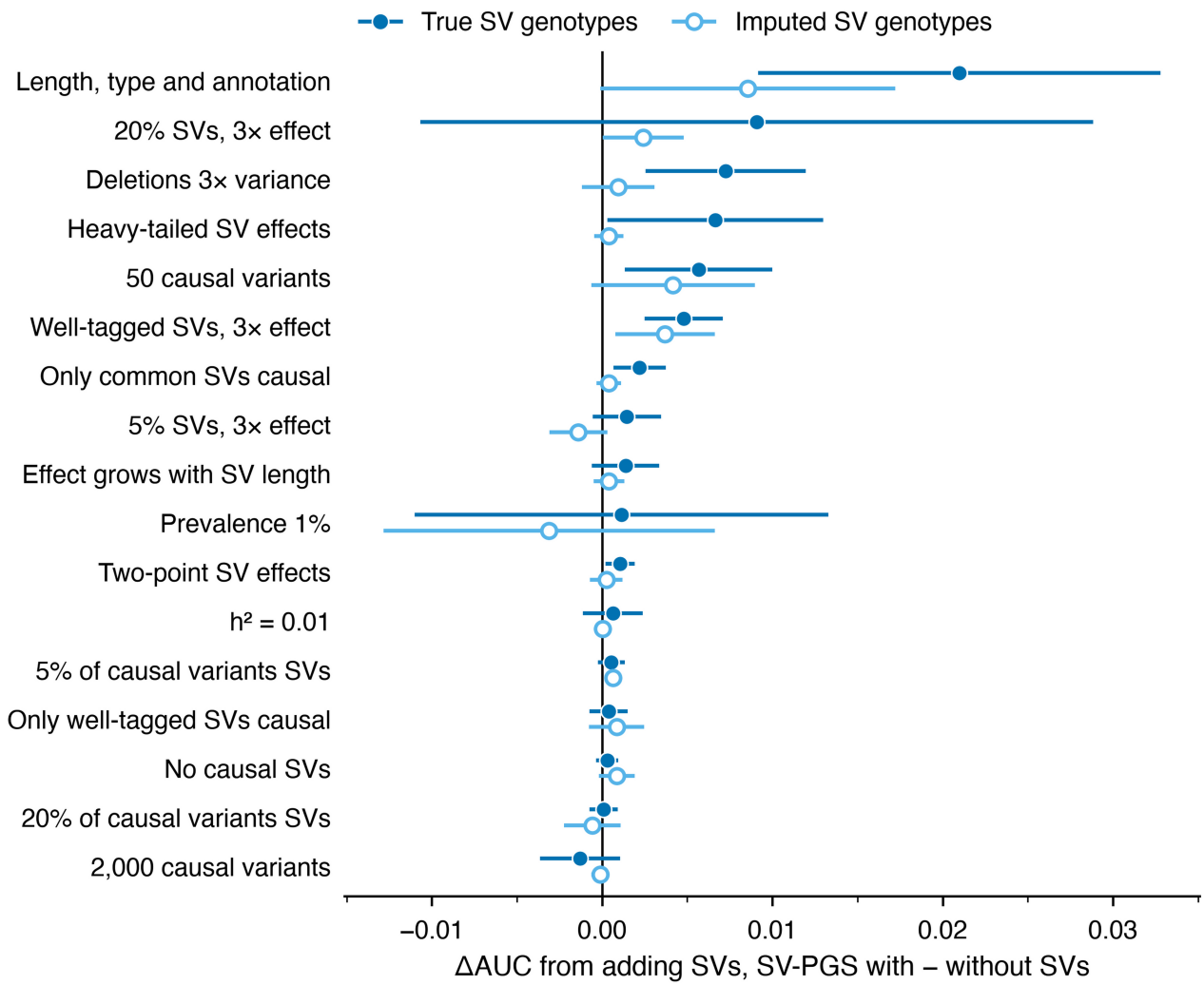


Figure 3 | SVs add accuracy when they are measured well; imputation removes most of the gain. Paired difference in test AUC between SV-PGS fitted with and without SVs, when every SV is measured by its true genotype (dark) or imputed by Beagle from SNV haplotypes (light), on one 3 Mb chromosome 22 window (60,000 records). Means with Student-t 95% intervals over replicates, training sizes 5,000-40,000 pooled. The true-genotype interval excludes zero for 7 of 17 architectures. Same simulated people, truth and fits; only the SV genotypes differ.

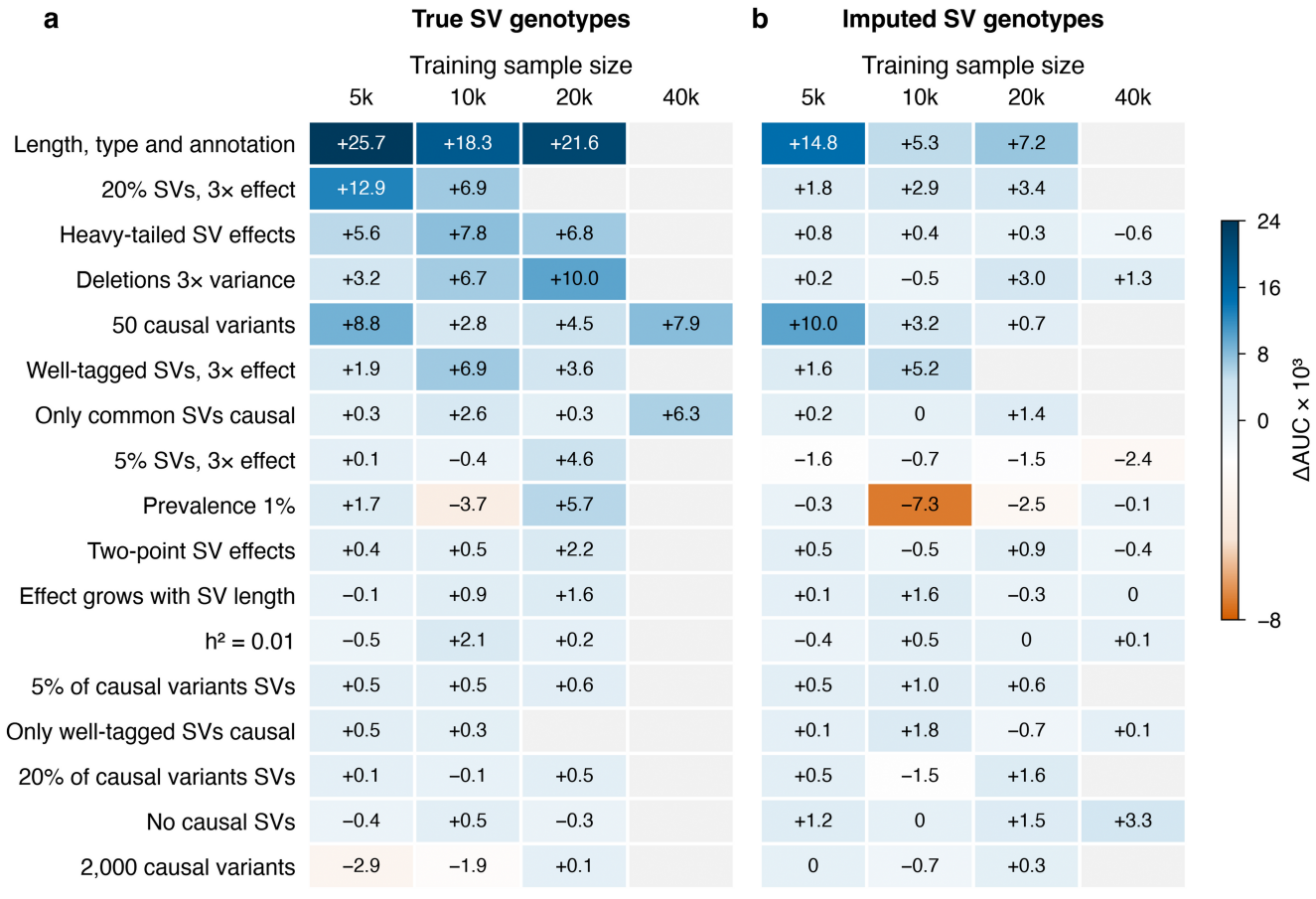


Figure 4 | Gain from adding SVs by architecture and training size. Mean paired ΔAUC (× 1,000) between SV-PGS fitted with and without SVs on one chromosome 22 window, with SVs measured by their true genotypes (a) or imputed (b). Grey cells had fewer than two replicates. Rows are ordered by the mean true-genotype gain.

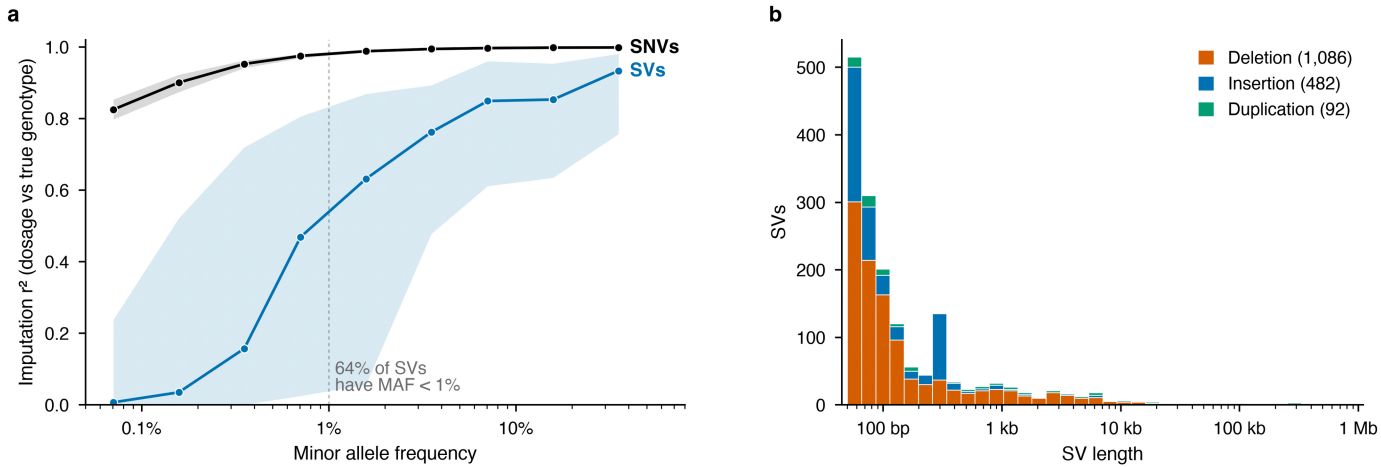


Figure 5 | The simulated SVs are mostly rare and short, and rare SVs impute poorly. a, Dosage r² between Beagle-imputed and true genotypes of 1,662 SVs and 6,000 sampled SNVs on chromosome 22 in 50,000 simulated people, by minor allele frequency: medians (points) and interquartile ranges (bands) per frequency bin. b, Lengths of the 1,662 measured SVs, stacked by class (sequence-resolved records without a type are classed by the sign of their length change). bench-sim v7 cohort from public 1000 Genomes haplotypes.

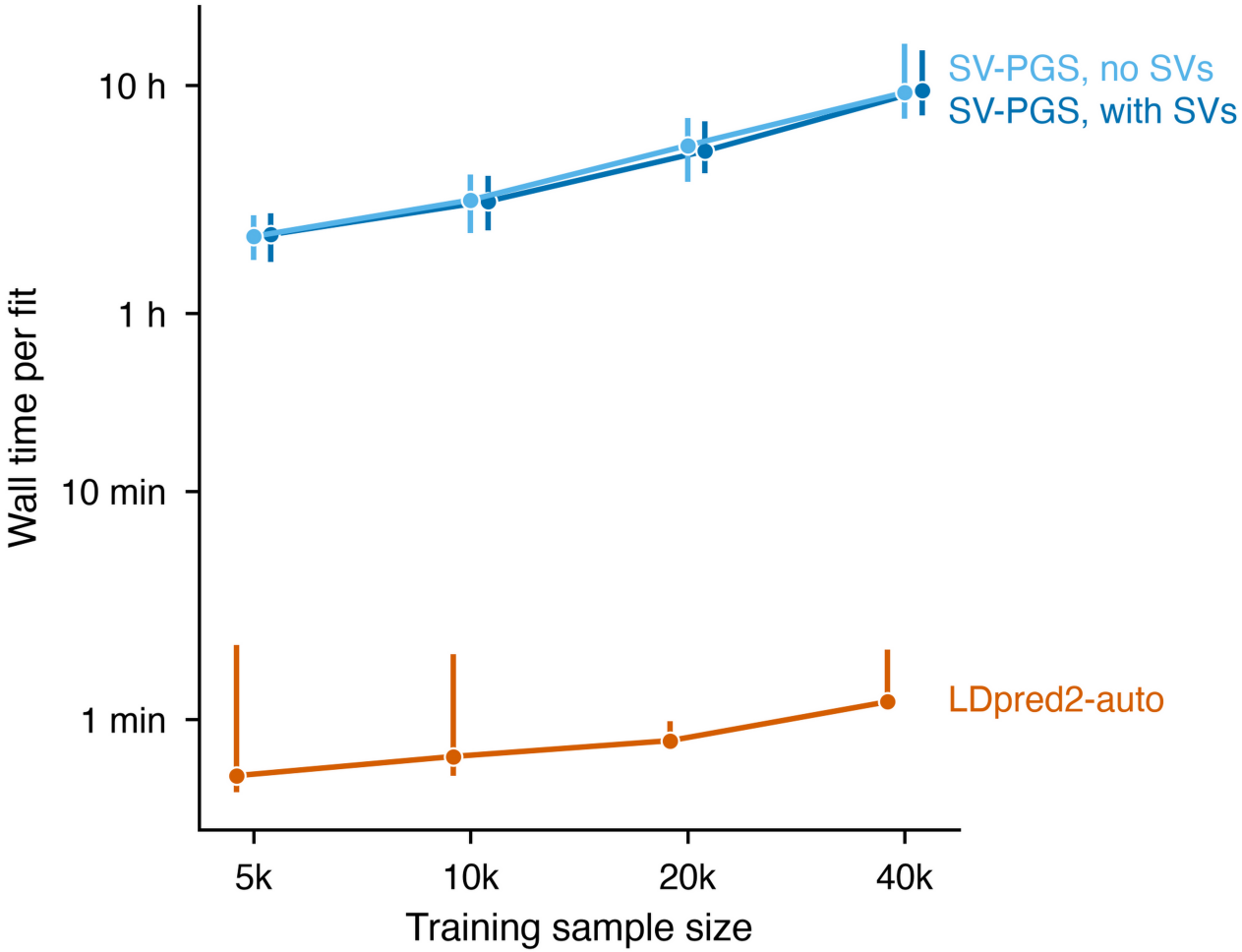


Figure 6 | Fit time. Wall time of one fit on chromosome 22 per method and training size, pooled over every architecture: median (points) and interquartile range (bars). MSI Slurm jobs, one fit per job; SV-PGS fits every variant by empirical Bayes, LDpred2-auto HapMap3-density SNVs.